20

Controlled- or modified-atmosphere storage and packaging

A reduction in the concentration of oxygen and/or an increase in carbon dioxide concentration of the storage atmosphere surrounding a food reduces the rate of respiration of fresh fruits and vegetables and also inhibits microbial and insect growth. When combined with chilling (Chapter 19), modified or controlled atmospheres are an increasingly important method of maintaining high quality in processed foods during an extended shelf life. Modified atmospheres are often included with other minimal processing methods (Chapters 1 and 9) as an important area of future development of mild processed, convenient and ready-to-eat foods that have good nutritional properties and a 'natural' image.

There remain differences in, and some confusion over, the terminology used. *Modified-atmosphere* storage (MAS) and packaging (MAP) are the use of gases to replace air around non-respiring foods without further controls after storage or packing. In *controlled-atmosphere* storage (CAS) and packaging (CAP), the composition of gas around respiring foods is monitored and constantly controlled, but with advances in 'active' packaging systems (Section 20.2.4) the distinction between MAP and CAP is no longer clear. In this book, MAP is used to refer to all methods to change atmospheres in packed food regardless of whether or not the atmosphere changes over time. It also includes vacuum packing (VP), equilibrium-modified-atmosphere (EMA) packaging, passive atmosphere modification (PAM), vacuum-skin packaging (VSP) and gas-exchange preservation (GEP). For definitions of these terms, see Section 20.2 or Church (1994).

In commercial operation, controlled-atmosphere storage (CAS) and modified-atmosphere storage (MAS) are mostly used with apples and smaller quantities of pears and cabbage. Modified-atmosphere packaging (MAP) is used for fresh foods and an increasing number of mildly processed foods, and is gaining in popularity as new applications are developed. Examples of MAP products include raw or cooked meats, poultry, fish, seafood, vegetables, fresh pasta, cheese, bakery products, sandwiches, sous vide foods, potato crisps, coffee and tea (Davies, 1995), and with new products including prepared salads, part-baked bread, croissants, pizzas, peeled fruits and prepared vegetables with dressing (Church, 1994).

The normal composition of air is 78% nitrogen and 21% oxygen, with the balance made up of carbon dioxide (0.035%), other gases and water vapour. An increase in the proportion of carbon dioxide and/or a reduction in the proportion of oxygen within specified limits (Section 20.2) maintains the original product quality and extends the product shelf life. This is achieved by:

- inhibiting bacterial and mould growth
- protecting against insect infestation
- reducing moisture loss
- reducing oxidative changes
- controlling biochemical and enzymic activity to slow down senescence and ripening.

CO_2 inhibits microbial activity in two ways: it dissolves in water in the food to form mild carbonic acid and thus lowers the pH of the product; and it has negative effects on enzymic and biochemical activities in cells of both foods and micro-organisms. However, close control over the degree of atmospheric modification is necessary to prevent physiological disorders in the living tissues and secondary spoilage by anaerobic micro-organisms in non-respiring foods. The effects of CO_2 on microbial growth are discussed by Dixon and Kell (1989) and reviewed by Farber (1991).

20.1 Modified- and controlled-atmosphere storage (MAS and CAS)

In MAS, the store is made airtight, and respiratory activity of fresh foods is allowed to change the atmosphere as oxygen is used up and CO_2 is produced (Chapter 19). In CAS, the concentrations of oxygen, carbon dioxide and sometimes ethylene (ethene) are monitored and regulated. Oxygen concentrations as low as 0%, and carbon dioxide concentrations of 20% or higher can be produced in for example grain storage, where these conditions destroy insects and inhibit mould growth. When storing fruits, a higher oxygen concentration is needed to prevent anaerobic respiration which would risk producing alcoholic off-flavours. Different types of fruit, and even different cultivars of the same species, require different atmospheres for successful storage and each therefore needs to be independently assessed. Examples of atmospheres for apple storage are 8% CO_2, 13% O_2 and 79% N_2 for Bramley's Seedling; and 5% CO_2, 3% O_2 and 92% N_2 for Cox's Orange Pippin at 3.5ºC to produce an increase from 3 months storage in air to 5 months under CAS. This can be further increased to 8 months using a CAS atmosphere of 1% CO_2, 1% O_2 and 98% N_2 although such low oxygen levels risk anaerobic respiration in other fruits. Refrigerated storage of winter white cabbage in 5% CO_2, 3% O_2 and 92% N_2 enables the crop to be stored until the following summer (Brennan *et al*., 1990). Details of the atmospheric composition required for other products, building construction, equipment and operating conditions are described by Ryall and Lipton (1979). Safety measures for operators when using increased concentrations of CO_2 are described in Chapter 19 and by Anon. (undated).

Storage is achieved using gas-tight stores, sealed using metal cladding and carefully sealed doorways. When CO_2 and oxygen levels change due to respiration in MAS, or when adjustment to atmospheric composition is needed in CAS, solid or liquid CO_2 can be used to increase gas concentration; controlled ventilation is used to admit oxygen or 'scrubbers' may be used to remove CO_2 and thus maintain a constant gas composition in the atmosphere. CO_2 scrubbers operate either by passing the atmosphere from the store over bags of hydrated calcium hydroxide (lime), under sprays of sodium hydroxide or

over activated carbon, to absorb the CO_2. Individual gases may be added from pressurised cylinders in MAS stores that are not completely gas-tight, to speed up the creation of the required atmosphere rather than relying on the action of the fruit alone.

The CO_2 content in the atmosphere can be monitored using sensors to measure differences in the thermal conductivity between CO_2 (0.015 W m^{-1} K^{-1}) and N_2 (0.024 W m^{-1} K^{-1}) and O_2 (0.025 W m^{-1} K^{-1}) or by differences in infrared absorption. Gas composition is automatically controlled by microprocessors using information from the sensors (Chapter 2) to control air vents and gas scrubbers, to maintain a pre-determined atmosphere.

MAS and CAS are useful for crops that ripen after harvest, or deteriorate quickly; even at optimum storage temperatures CA stores have a higher relative humidity (90–95%) than normal cold stores and therefore retain the crispness of fresh foods and reduce weight losses.

The main disadvantages of MAS and CAS are economic: crops other than apples (and to a lesser extent cabbage and pears) have insufficient sales to justify the investment. Short season crops, which increase in price out of season, justify the additional costs of MAS or CAS, but the plant cannot be used throughout the year. Also plant utilisation cannot be increased by storing crops together, because of the different requirements for gas composition, and the risk of odour transfer. Other limitations of MAS and CAS are as follows:

- the low levels of oxygen, or high levels of carbon dioxide, which are needed to inhibit bacteria or fungi, are harmful to many foods
- CAS conditions may lead to an increase in the concentration of ethylene in the atmosphere and accelerate ripening and the formation of physiological defects
- an incorrect gas composition may change the biochemical activity of tissues, leading to development of off-odours, off-flavours, a reduction in characteristic flavours, or anaerobic respiration

Table 20.1 Maximum levels of carbon dioxide and minimum levels of oxygen for storage of selected fruits and vegetables

Food	CO_2 (%)	O_2 (%)
Apple[a] (Golden Delicious)	2	2
Asparagus (5ºC)	10	10
Avocado	5	3
Banana	5	–
Broccoli	15	1
Cabbage	5	2
Carrot	4	3
Cauliflower	5	2
Citrus fruits	–	5
Cucumber	10	3
Lettuce	1	2
Onion	10	1
Pea	7	5
Pear (Bartlett)	5	2
Potato	10	10
Spinach	20	–
Strawberry	20	2
Sweetcorn	20	–
Tomato	2	3

[a] Dewey (1983) describes gas compositions for other varieties of UK apples.
From Anon. (1979) and Ryall and Pentzner (1982).

- tolerance to low oxygen and high carbon dioxide concentrations (Table 20.1), varies according to type of crop, conditions under which a crop is grown and maturity at harvest
- different cultivars of the same species respond differently to a given gas composition, and growers who regularly change cultivars are unwilling to risk losses due to incorrect CAS conditions
- economic viability may be unfavourable owing to competition from other producing areas which have different harvest seasons, and higher costs of CAS over a longer storage period (twice that of cold storage).

An alternative approach is storage in a partial vacuum which reduces the oxygen concentration by the same proportion as the reduction in air pressure (that is, if the pressure is reduced by a factor of 10, then the oxygen concentration is reduced by the same factor). The main advantages are the continuous removal of ethylene and other volatiles from the atmosphere and precise control of air pressure (±0.1%). However, the method is not commonly used owing to the higher costs.

20.2 Modified-atmosphere packaging

20.2.1 MAP for fresh foods

MAP (or gas flushing) is the introduction of an atmosphere, other than air, into a food package without further modification or control (Wilbrandt, 1992). Although the term 'MAP' is used throughout this book to describe packaging in modified atmospheres, other terminology is in use to more specifically designate different operations, including *controlled-atmosphere packaging* (CAP) (continuous monitoring and control of gas composition in bulk containers), *equilibrium-modified atmosphere* (EMA) or *passive atmosphere modification* (PAM) (gas flushing of packs of fresh fruits or vegetables or sealing without gas modification to allow a gas equilibrium to be established as a result of respiration), *vacuum packing* (VP) (the removal of the majority of air from a pack that has low oxygen permeability, with subsequent changes in gas composition due to metabolic activities of the product or micro-organisms), *vacuum-skin packaging* (VSP) (placing a softened film over the product and applying a vacuum to form a skin) and *gas-exchange preservation* (GEP) (replacing air with a series of gases in quick succession to inhibit enzymes or kill micro-organisms, before packing in nitrogen) (Church, 1994; Davies, 1995).

MAP is used to extend a product shelf life to give processors additional time to sell the food without sacrificing quality or freshness (Table 20.2). The potential advantages and limitations of MAP are shown in Table 20.3 and differences in the market potential for MAP foods in Europe and USA are reviewed by Davies (1995). The atmosphere is not, however, constant in all MAP products and will change according to:

- the permeability of the packaging material
- microbiological activity
- respiration by the food.

Successful MAP requires raw materials with a low microbiological count and strict temperature control throughout the process (Chapter 19). The three main gases used in MAP are nitrogen, oxygen and CO_2, although others, including carbon monoxide, nitrous oxide, argon, helium and chlorine have also been investigated, but largely eliminated due

Table 20.2 Extension of shelf life using MAP

Product	Shelf life (days)	
	Air	MAP
Beef[a]	4	12
Bread[b]	7	21
Cake[b]	14	180
Chicken[a]	6	18
Coffee[b]	3	548
Cooked meats[a]	7	28
Fish[a]	2	10
Fresh pasta[a]	2	28
Fresh pizza[a]	6	21
Pork[a]	4	9
Sandwiches[a]	2	21

[a] Refrigerated storage.
[b] Ambient storage.
Adapted from Brody (1990) and Blackistone (1998b).

to safety, cost or effects on food quality. Nitrogen is inert and tasteless, with low solubility in both water and fats. It is used to replace oxygen and thus inhibit oxidation or the growth of aerobic micro-organisms.

Oxygen is used in MAP to maintain the red colour of oxymyoglobin in unprocessed meats, or to permit respiration of fresh produce, but in other applications its level is reduced to prevent growth of spoilage micro-organisms and oxidative rancidity. Typically, the shelf life of fresh red meat is extended from 3 days to 7 days at 0–2ºC by packaging in an 80% O_2 / 20% CO_2 atmosphere, but this may cause problems of oxidative rancidity in fatty fish or development of off-colours in cured meats. Bacon, for example, is therefore packed in 35% O_2 / 65% CO_2 or 69% O_2 / 20% CO_2 / 11% N_2. In both atmospheres the oxygen concentration is sufficient to inhibit anaerobic bacteria. Pork, poultry and cooked meats have no oxygen requirement to maintain the colour, and a higher carbon dioxide concentration (90%) is possible to extend the shelf life to 11 days. Further details are given by Blakistone (1998a).

In fresh fruits and vegetables, a concentration of 10–15% carbon dioxide is required to control decay. Some crops can tolerate this level (for example strawberries and spinach) but most cannot (Table 20.1) and MAP is unsuitable. A high carbon dioxide

Table 20.3 Advantages and limitations of MAP

Advantages	Limitations
Increased shelf life of 50–400%	Added cost
Extended storage results in reduced economic losses and wider distribution radius	Temperature control required
	Different gas compositions for each type of product
Fewer distribution deliveries leads to lower costs	Requirement for special equipment and operator training
Little or no need for chemical preservatives	Increased pack volume has impact on transport and retail display costs
Easier separation of sliced foods (except vacuum packing)	Benefits are lost once the pack is opened or leaks
Good presentation of products	Product safety to be established for some foods

Adapted from Davies (1995), Farber (1991) and Blakistone (1998b).

concentration prevents mould growth in cakes and increases the shelf life to 3–6 months. Other bakery products (for example hamburger buns) have the shelf life increased from 2 days to 3–4 weeks (Guise, 1983).

CO_2 dissolves in both water and fats in a food and is more soluble in cold water than it is in warm water. It is absorbed into fish tissue, which lowers the pH and increases drip losses. In MAP, the absorption of CO_2 should be carefully controlled to prevent too great a reduction in gas pressure which causes collapse of the pack. Nitrogen is often added as a filler gas to prevent pack collapse, although in some products collapse may be advantageous (for example hard cheeses), where a tight pack is formed around the product. Additionally, the relative volume of gas and product is important to ensure the effectiveness of MAP (a sufficiently high gas:product ratio for the gas to have a preservative effect). There should therefore be adequate space between the product and the package to contain the correct amount of gas.

For fresh produce, the aim of MAP is to minimise respiration and senescence without causing damage to metabolic activity that would result in loss of quality (Section 20.1). However, the effects of low oxygen and raised CO_2 concentrations on respiration are cumulative, and respiration also continually alters the atmosphere in a MA pack. The rate at which oxygen is used up and CO_2 is produced also depends on the storage temperature. The optimum gas composition in a pack is therefore difficult to predict or achieve. In practice, the CO_2 concentration is increased by gas flushing before sealing and a film that is permeable to oxygen and CO_2 is selected to enable respiration to continue (see also Section 20.2.4). Changes in gas composition during storage depend on.

- the respiration rate of fresh foods, and hence the temperature of storage
- the permeability of the packaging material to water vapour and gases
- the external relative humidity, which affects the permeability of some films
- the surface area of the pack in relation to the amount of food it contains.

MAP permits an extension to the shelf life of cut red meat of up to 18 days, and for ground beef up to 10 days. Cut lettuce has a two-week shelf life at 0–1.1°C (Brody, 1990) (Table 20.2). Details of MAP for fresh produce are given by Garrett (1998).

20.2.2 MAP for processed foods

For processed (that is non-respiring) foods, atmospheres should be as low as possible in oxygen and as high as possible in CO_2 without causing the pack to collapse or produce changes to the flavour or appearance of the product. Ground coffee, for example, is protected against oxidation by MAP using a CO_2/N_2 mixture or by vacuum packing.

Reducing the concentration of oxygen inhibits the development of 'normal' spoilage micro-organisms, especially *Pseudomonas* sp. (Walker, 1992). Other spoilage bacteria that can grow in low oxygen concentrations grow more slowly and so extend the time taken for spoilage to occur, for example lactic acid bacteria or *Brochothrix thermosphacta,* which cause spoilage by souring (Nychas and Arkoudelos, 1990). Concern has been expressed over potential risks to consumer safety from modified atmospheres or vacuum packaging because they inhibit 'normal' spoilage micro-organisms and thus allow food to appear fresh, while permitting the growth of anaerobic pathogens. Details of pathogens found on chilled foods are given in Chapter 19. Several pathogens including *Clostridium botulinum*, *Listeria monocytogenes*, *Yersinia enterocolitica*, *Salmonella* sp., and *Aeromonas hydrophila* are anaerobes or facultative anaerobes (Blakistone, 1998b). A large number of studies of the effect of MAP on the

microbiology of foods are reported; for example meat poultry and fish (Church, 1998; Finne, 1982; Christopher *et al.*, 1980), baked goods (Knorr and Tomlins, 1985; Ooraikul, 1982). These are reviewed for example by Davies (1995), Church (1994), Ooraikul and Stiles (1991) and Farber (1991). The studies have indicated that growth of pathogens in MAP products is no greater, and frequently lower than in aerobically stored foods. However, for products in which there is a potential safety hazard, it is recommended that one or more of the following criteria are met:

- water activity (Chapter 1) is below 0.92
- pH is below 4.5
- use of sodium nitrite or other preservative
- the temperature is maintained below +3ºC.

The application of HACCP techniques (Chapter 1) also plays a major role in ensuring the safety of all MAP foods. Different foods respond in different and sometimes unpredictable ways to modified atmospheres, and each product should therefore be individually assessed using MAP trials, to monitor microbial activity, moisture content, pH, texture, flavour and colour changes in order to determine the optimum gas composition. Care is also needed to prevent temperature abuse during processing and distribution, and high standards of hygiene should be used throughout the production process (also Chapter 19).

Examples of gas mixtures that are used for fresh and processed foods are shown in Table 20.4. In MAP of bread, CO_2 inhibits mould growth and the retention of moisture maintains softness. This is not inhibition of staling (a process that involves partially reversible crystallisation of starch), but the effects are similar. Spraying bread with 1% ethanol doubles the ambient shelf life, by retarding mould growth and an apparent inhibition of staling (also Section 20.2.4). A novel MAP approach to packing baguettes is to pack them straight from the oven while the CO_2 produced by the fermentation is still being emitted. As they are placed into thermoformed packs the CO_2 expels air and

Table 20.4 Gas mixtures used for selected MAP foods

Commodity	% CO_2	% O_2	% N_2
Baked products	60	0	40
Cheese (hard)	60	0	40
Cheese (mould ripened)	0	0	100
Cream	0	0	100
Crustaceans	40–60	20–30	0–30
Dry snackfoods	20–30	0	70–80
Fish (oily)	30–60	0	40–70
Fish (white)	40–60	20–30	0–30
Fruit/vegetables	3–10	2–10	80–95
Kebabs	40–60	0–10	40–60
Meat (cooked)	25–30	–	70–75
Meat (cured)	20–35	–	65–80
Meat (red)	15–40	60–85	0–10
Meat pies	20–50	–	50–80
Pasta (fresh)	50–80	–	20–50
Pizza	40–60	0–10	40–60
Poultry	20–50	–	50–80
Quiche	40–60	–	40–60
Sausage	60	40	0

Adapted from Day (1992) and Smith *et al*. (1990).

saturates the atmosphere to give 3 month shelf life at ambient temperature. The consumer briefly heats the bread in an oven to create a crust and produce a product that is similar to freshly baked bread (Brody, 1990).

20.2.3 Packaging materials for MAP

The two most important technical parameters of packs for MAP are gas permeability and moisture vapour permeability. Packaging materials are classified according to their barrier properties to oxygen into:

- low barrier (>300 cc m^{-2}) for over-wraps on fresh meat or other applications where oxygen transmission is desirable
- medium barrier (50–300 cc m^{-2})
- high barrier (10–50 cc m^{-2})
- ultra high barrier (<10 cc m^{-2}), which protect the product from oxygen to the end of its expected shelf life.

Typical film materials are single or coextruded films or laminates of ethylene vinyl alcohol (EVOH), polyvinyl dichloride (PVDC), polyethylene terephthalate (PET), polypropylene (PP), polyethylene (PE), polyester, amorphous nylon (polyamide resin) and nylons, although the last provides only moderate barrier. Details of types of film and their permeability to moisture and gases are given in Chapter 24 and described by Greengrass (1998). Films are usually coated on the inside of the pack with an antifogging agent, typically a silicone or stearate material, to disperse droplets of condensed moisture and permit the food to be visible. New developments include films that change permeability to moisture and gases under specified temperature conditions that are designed to match the respiration rate of a fresh product (Vermeiren *et al.*, 1999 and Chapter 24).

In MAP operation, air is removed from the pack and replaced with a controlled mixture of gases, and the package is heat sealed. In batch equipment, pre-formed bags are filled, evacuated, gas flushed and heat sealed in a microprocessor-controlled programmed sequence. In continuous operation, food is packaged in three basic ways: in semi-rigid, thermoformed trays covered with film that has the required permeability (for example, for meats); or second, in pillow pouches (for example, for fresh salads). The design of MAP packs for fresh produce is described by Yam and Lee (1995). Third, foods such as baked products are packed in horizontal form-fill-seal equipment or 'flowpacks'. All types allow space around food for the gas. The different types of packaging systems are described in detail by Hastings (1998) and in Chapter 25.

20.2.4 Active packaging systems

The development of active packaging systems (also termed 'intelligent' packaging or 'smart films'), is a significant new area of MAP technology (Table 20.5). They have the following capabilities:

- edible moisture barriers for fresh fruits and vegetables or edible oxygen barriers to prevent enzymic browning
- ethylene scavengers – these are sachets of silica gel containing potassium permanganate, which oxidises ethylene to slow the ripening of fruits
- oxygen scavengers to create low-oxygen atmospheres or slow the oxidation of lipids

Table 20.5 Examples of active packaging systems

Method	Variations	Examples of products
Oxygen scavenger	Powdered iron oxide Ferrous carbonate Iron/sulphur Platinum catalyst Glucose-oxidase enzyme Alcohol oxidase enzyme	Cookies, cured meats, pizza crusts, bread, rice cakes
Carbon dioxide scavenger/ emitter	Powdered iron oxide/calcium hydroxide Ferrous carbonate/metal halide	Coffee, fresh meats/fish
Preservative	BHA/BHT (Appendix C) Sorbates Mercurial compounds Zeolite system	Meats, fish, bread, cereals, cheese
Ethanol emitter	Ethanol spray Encapsulated ethanol	Cakes, bread, buns, tarts, fish
Moisture absorber	PVA blanket	Fish, meats, poultry
Temperature or humidity control	Non-woven plastics PET containers Foams	Prepared entrées, meats, poultry, fish

From Day (1992).

and films that scavenge off-odours or carbon dioxide. Oxygen scavenging sachets contain iron which is oxidised in the presence of water vapour to produce ferric hydroxide. If the oxidation rate of the food and the oxygen permeability of the film are known, the amount of iron needed in the sachet for the required shelf life can be calculated. Other approaches include a film that contains a reactive dye and ascorbic acid; a film incorporating platinum to reduce oxygen to water vapour; and attachment of immobilised enzymes, including glucose oxidase and alcohol oxidase to the inner surface of a film. The products of these enzymic reactions also lower the surface pH of the food and release hydrogen peroxide which extend the shelf life of fresh fish. Others include a film that contains an organic chelation agent that binds oxygen, and a film that incorporates a free-radical scavenger to react with oxygen. A sachet containing iron powder and calcium hydroxide scavenges both oxygen and CO_2 and has been used to produce a threefold extension to the shelf life of packaged ground coffee. Conversely in some CAP/MAP applications, high levels of CO_2 are required, but many films are 3–5 times more permeable to CO_2 than to oxygen. In these situations, a carbon dioxide generator is used. In other situations, low oxygen levels can create favourable conditions for the growth of pathogenic anaerobic bacteria and a 'smart' film which permits a substantial increase in gas permeability with higher temperatures, is used to re-oxygenate packs of food to prevent anaerobic conditions from forming.

- zeolite films to inactivate micro-organisms on food surfaces and sachets and films that release microbial inhibitors
- ethanol that is trapped in silica gel, contained in a sachet made from a film that is highly permeable to ethanol vapour, has been used to extend the shelf life of bakery products, cheese and semi-dried fish products. Similarly a sulphur dioxide generating film or a film that releases trapped sorbate have been used to extend the shelf life of grapes by preventing mould growth. A sachet system which rapidly increases absorption of moisture as the temperature approaches the dew point is used to prevent

droplets of water forming on the product which could promote microbial growth. A similar effect is produced by trapping propylene glycol or diatomaceous earth in a film placed in contact with the surface of fresh meat or fish to absorb water and injure spoilage bacteria.

These developments are described by Labuza and Breene (1989), Church (1994) and Smith *et al*. (1990). Other developments, including films that have selective gas transmission (by tailoring the film materials or by microperforation), selective water vapour transmission, the use of noble gases and films that change permeability to compensate for temperature fluctuations are described by Gorris and Peppelenbos (1999).

Oxygen scavengers are the most widely developed application to date and operate in two ways to remove oxygen from a pack: either small amounts of chemicals are placed in a sachet contained within the food pack; or foods are packed in oxygen-scavenging films, which absorb oxygen from the headspace above the food. In the first method, sachets of ferrous powders or similar chemicals that can absorb large amounts of oxygen are used (Table 20.5). The use of oxygen-scavenging chemicals is widespread in Japan, but has not been widely accepted in Europe or USA to date, possibly because of fears over accidental consumption of the chemicals or litigation if they are consumed. The use of oxygen-absorbing labels (Anon., 1994) or sachets contained in sealed compartments in a pack may overcome this resistance. Immobilisation of oxidising enzymes (glucose oxidase, alcohol oxidase) on the inner surface of films has also been shown to be feasible, but is too expensive at present. Applications of oxygen scavengers so far include bakery products, pre-cooked pasta, cured and smoked meats, cheese, spices, nuts, coffee (Davies, 1995), jelly confectionery, soybean cakes, rice cakes, soft cakes and seaweed-based foods in oriental countries (Table 20.5).

Systems for CO_2 production involve placing sachets of chemicals in the base of a tray, covered by a plastic mesh. When activated by moisture or water vapour, the sachets either release CO_2 or in other applications, they absorb ethylene and/or CO_2, depending on the chemicals used. Ethylene absorption delays ripening (Chapter 19) and systems based on activated carbon or potassium permanganate have been developed. Other systems include combined oxygen and CO_2 scavenging in packed, freshly roasted coffee beans, one-way valves which release CO_2 from the pack without allowing other gases to enter (for mould-ripened cheese), and high CO_2 permeable films for coffee (Church, 1994). Japanese companies have also developed an oxygen-sensitive ink and an indicator that changes from pink to blue when oxygen levels rise from $<0.1\%$ to $>0.5\%$ (Church, 1994), which are used to ensure that gas composition is maintained and may have applications to check non-destructively pack integrity.

Ethanol has anti-microbial properties, especially against moulds, and ethanol generators have been used to increase the shelf life of baked products, cheeses and semi-dried fish.

The growing awareness of environmental problems caused by packaging materials has renewed interest in edible protective superficial layers (EPSL). These are applied directly to the surface of a food and act as an additional hurdle to loss of quality and protection against microbial spoilage. Active EPSLs, with antimicrobial properties (for example using sorbic acid) or antioxidant properties have been developed to fix the additives at the product surface where they are required, and therefore reduce the amounts that are used. Flexible, hydrophilic EPSLs, having good resistance to breakage and abrasion have been developed from gluten and pectin (Gontard *et al*., 1992, 1993). Developments in active packaging are reviewed by Vermeiren *et al*. (1999).

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